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Boobphachard Chansawang,¹ Rahat Zarin,¹ Usa Wannasingha Humphries,^{1,a)} Prungchan Wongwises,² Muhammad Waqas,² and Angkool Wangwongchai¹

AFFILIATIONS

¹ Department of Mathematics, King Mongkut's University of Technology Thonburi, Bangkok, Thailand

² The Joint Graduate School of Energy and Environment, King Mongkut's University of Technology Thonburi, Bangkok, Thailand

^{a)} Author to whom correspondence should be addressed: usa.wan@kmutt.ac.th

ABSTRACT

Geophysical domains typically exhibit intricate, irregular boundaries characterized by fractal-like geometries, while underlying physical processes operate across a broad spectrum of spatial scales. The challenge lies in generating spatial discretization of these domains that conform to their geographical constraints, utilizing anisotropic, fully adaptive meshes. This problem is compounded by the vast range of scales and a notably heterogeneous parameter space. Current methodologies often rely on ad hoc, model-specific, or application-dependent approaches, which lack comprehensive descriptions. Consequently, the development of new spatial domains is labor-intensive, prone to errors, challenging to replicate, and difficult to maintain consistency due to substantial human involvement. This predicament poses obstacles to the reproducibility of simulations and the establishment of provenance in data handling and model initialization, and it hinders rigorous model intercomparisons. Furthermore, the likelihood of discrepancies in model initialization and forcing parameters increases when employing flexible adaptive meshes. This paper introduces a systematic approach to the automated generation of adaptive meshes for geophysical models. This method is efficient in its generation process and readily reproducible, offering robust and consistent adherence to the source data. The proposed approach facilitates research in complex multi-scale geophysical domains, which would be challenging using existing methods. A simulation of monthly mean currents was carried out as a case study in the Gulf of Thailand. Results revealed that the simulated current circulations agreed with the observation. Examples of its application in various ongoing geophysical modeling endeavors illustrate its effectiveness.

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I. INTRODUCTION

The utilization of numerical simulation has emerged as a crucial method for addressing intricate challenges in both engineering and scientific domains. Leveraging the growing computational power, this contemporary numerical approach directly tackles complex problems, delving into the intricacies of the original scenario without heavily relying on assumptions. In contrast to the conventional theoretical practice, which often resorts to simplifications and approximations, this modern approach aims to comprehensively engage with the complexities of the studied problems, enabled by the advancements in computer capabilities.¹ In recent years, substantial

progress has been made in developing numerical simulation models tailored for geophysical processes. These models employ finite difference methods on a transformed grid, incorporating the benefits of unstructured models. While unstructured models traditionally use finite element or finite volume methods, our approach adapts finite differences in conjunction with grid transformations to emulate the advantages of unstructured models. It is essential to recognize that this study focuses on harnessing the flexibility and adaptability of unstructured mesh techniques without adhering strictly to the traditional numerical methods associated with unstructured models.² The Finite-Volume Community Ocean Model (FVCOM) is renowned for its focus on coastal-scale modeling, contributing to

a more detailed understanding of coastal dynamics and processes.¹ D-flow^{2,3} and the Second-generation Louvain-la-Neuve Ice-Ocean Model (SLIM)⁴ have proven instrumental in simulating flow patterns and processes in rivers and marine estuaries, aiding in managing and preserving these critical environments. QUODDY has been deployed in the complex Canadian Arctic Archipelago, facilitating investigations into the intricate dynamics of this challenging region.⁵ T-UGOm is recognized for providing improved tidal statistics, enabling a more comprehensive understanding of tidal behaviors.⁶ Fluidity is utilized for flexible non-hydrostatic studies, offering the capability to explore a wide range of geophysical phenomena with adaptability and precision.⁷ The finite-element sea ice-ocean model (FESOM) was recently integrated into large-scale,⁸ internationally coordinated climate model intercomparisons, such as CMIP⁹ and CORE,¹⁰ marking a significant stride in coupled climate and ocean modeling research. These current approaches, harnessing unstructured mesh methods in ocean modeling, represent a pivotal development in geophysical simulations. However, for a more in-depth understanding of these approaches' intricate nuances and comparative analysis, one can refer to the comprehensive exploration provided by Ref. 11.

In geophysical modeling, unstructured mesh approaches are superior to regular, structuring grids in many ways. Their remarkable ability to conform to complex, fractal-like boundaries, often distinctive features of geophysical domains such as ocean coastlines and bathymetry, distinguishes them. Thompson *et al.* developed the adaptive mesh coordinates, in which the transformation fits to the boundary. The grids were suitable for calculating the potential flow of airfoils and other applications.¹² According to authors such as Danilov, these flexible, conforming boundaries are crucial for accurately representing local features that significantly influence ocean dynamics.¹³ Because of their flexibility, unstructured mesh techniques can offer a more accurate depiction of geophysical processes. Moreover, the fundamental support of variable spatial resolutions provided by unstructured mesh techniques is essential for ocean modeling. Works by Kantha and Clayson (e.g., Ref. 14) document the wide range of spatial and temporal scales over which oceanic processes operate. Large-scale circulation, latitude-driven variations such as Rossby effects, tidal phenomena, and smaller-scale processes such as internal and gravity waves, double diffusion, and oceanic turbulence encompass a range of phenomena. As (e.g., Ref. 15) has been shown, unstructured meshes make optimizing spatial resolution and effectively capturing this wide range of scales possible. They lessen the requirement for empirical parameterizations and nested grids by doing this.

Furthermore, when finer scales are added, unstructured methods may yield higher computational efficiency. The feasibility of geophysical simulations can be significantly impacted by a significant reduction in computational resource requirements, as estimated in Ref. 16. These reductions can range from 5 to 17 times less. Danilov pointed out that unstructured mesh models can be a good option in some situations, even though they might not completely replace structured modeling techniques.¹³ In these cases, accurate representation of complex geophysical phenomena is aided by the flexibility and effectiveness of unstructured approaches. The advancement of unstructured techniques for ocean modeling has been a focal point of research and a significant and continuing scientific discourse during the last fifteen years. The annual IMUM (Implicit Multi-scale

Unstructured Mesh) workshops are centered around this discussion, as reported in Ref. 17, which were also included in a special issue of Ocean Modeling. The main aim of these conversations has been to tackle different obstacles related to the application of unstructured models, specifically focusing on the discretization of the dynamic core. Unstructured gridded models are more complex regarding numerical discretization than their gridded counterparts. However, their intricacy goes beyond the numerical to include problems with model setup and initialization, a crucial mesh generation task. Unstructured gridded models require a heterogeneous array of data types and more sophisticated data processing procedures than regular gridded models, which rely on a simple land mask to define the domain boundaries and retain data in a structured array format. It poses another significant obstacle to their general adoption, especially outside the specialized community devoted to their creation and application.

Subdividing a continuous spatial domain into discrete elements is the current method for creating spatial discretization, as reviewed in detail by Ref. 18. It relates primarily to the formation of a discrete tessellation, or honeycomb pattern, in the context of this discussion; this is a generalized form of triangulation. These current approaches frequently have drawbacks, mainly because they are labor-intensive and difficult to duplicate because they are customized for models or applications. Moreover, they are usually created with a single goal: to test and progress the advancement of numerical discretization methods. One of these methods' significant drawbacks is that mesh discretization cannot be easily edited and adjusted. Unstructured mesh techniques have this fundamental advantage of flexibility. Furthermore, these methods' processes are often poorly documented, making them harder for beginners in the field to access. A general, easily implemented technique to formalize the description of spatial discretization is needed, given the increasing use of mature unstructured mesh numerical simulation models in the geophysical domain. This approach would help modelers realize the full potential of these sophisticated modeling techniques by making it easier to engage with model intercomparisons.

Creating an unstructured spatial discretization can be summed up as the main task. A set of self-consistent geospatial data fields should drive this deterministic and automated generation process, with a crucial focus on guaranteeing the preservation of self-consistency from the source data to the final spatial discretization. This problem occurs in the context of evolving geophysical simulations where the complexity of the domain bounds is increasing. According to predictions by Ref. 16, as simulations cover a more comprehensive range of scales, global models must include coastal scales as low as 1.5 km within the next ten years. The challenge is exacerbated by the need for a wide range of datasets, incorporating different physical processes and diversifying scales. The complexity of oceanic behavior necessitates adopting dynamically adaptive methods, wherein the mesh resolution is adjusted over time in response to evolving solution fields. For instance, as fronts or eddies shift positions, the mesh resolution should ideally adapt to track or anticipate these movements. Similarly, for sporadic events such as seasonal effects requiring heightened resolution at specific times, the mesh should have the flexibility to concentrate on the evolving process and de-emphasize as the phenomenon subsides. This dynamic adaptability proves essential in capturing and

representing the transient and varied nature of oceanic processes over extended simulation periods.^{7,19} As a result, given the complexity and variety of modern geophysical modeling, it becomes increasingly important to guarantee that the meshes and components utilized in constructing this spatial discretization maintain mutual consistency throughout.

This study's compelling focus on advanced numerical simulations using unstructured mesh techniques demonstrates their pivotal role in enhancing accuracy and efficiency in geophysical modeling. These techniques contribute significantly to understanding intricate oceanic behaviors by addressing complex challenges and promoting adaptability. The motivation lies in overcoming obstacles, such as model setup complexities, and ensuring mutual consistency in spatial discretization, exemplified through a case study on the Gulf of Thailand (GoT), where adaptive mesh coordinates prove essential for precise predictions and planning in marine-related industries. The GoT has an irregular shape;²⁰ in calculating nearshore ocean currents along the complicated coastline of the GoT, the regular mesh will cause inconsistency between the boundaries of the discrete and physical domains. An adaptive mesh is used to avoid the problem. Adaptive mesh coordinates are applied to predict the monthly mean current circulation in the GoT over the continental shelf near Thailand. The GoT's significance involves marine lines, the fishing industry, and tourist attractions along their coasts and on many islands. Studying the monthly mean current circulation in the GoT is essential in planning and developing related industries. In addition, the movement of pollutants caused by various sources and their effect within the area are an additional benefit. Therefore, the main objective of this research is to use two-dimensional oceanic models to study the current circulation over the GoT. The adaptive mesh was applied on Thailand's complicated coastline to eliminate computational errors on the boundary.

II. MATERIALS AND METHODS

A. Study area

The research investigation was conducted in the GoT situated in Southeast Asia (SEA), near the western region of the South China Sea (SCS), as illustrated in Fig. 1. The Gulf of Thailand is characterized as a partially enclosed water body, spanning ~400 km by 800 km and containing a total area of ~320 000 km².²¹ Geographically, it is positioned between 6° N–14° N and 99° E–105° E, bordered by Cambodia, Malaysia, Thailand, and the Socialist Republic of Vietnam.^{20,22} The GoT is integral to the Sunda Shelf, serving as an underwater linkage that connects SEA, Malaysia, Sumatra, Java, and Borneo. This region is marked by its relatively shallow depths, a distinctive feature of the Sunda Shelf. The geographical interconnection facilitated by the Sunda Shelf plays a significant role in shaping the hydrodynamic and ecological characteristics of the GoT and surrounding areas.^{23,24}

B. Initial data and quality checking

Table I presents meteorological data alongside their respective sources. The wind velocity data utilized in this study were sourced from the European Center for Medium-Range Weather Forecasts (ECMWF) 10 m above the sea surface. The climatological monthly

mean wind information is accessible at a specified grid resolution of $0.25^\circ \times 0.25^\circ$, while the 6-h wind velocity data are provided at a distinct grid resolution of $1.125^\circ \times 1.125^\circ$. These data points, expressed in meters per second (m/s), were extracted from the ECMWF dataset for analysis and subsequent use in the study.

The bottom topography for the GoT and AS has been sourced from the global Digital Bathymetric Data Base-5 min (DBDB5) binary data file. DBDB5 is a comprehensive digital bathymetric database offering ocean depth information with a 5-min longitude–latitude gridded resolution, as illustrated in Fig. 2. The Naval Oceanographic Office (NAVOCEANO) developed this dataset to support the generation of bathymetric chart products and provide bathymetric data to be integrated with other geophysical and environment parameters for ocean modeling. This study employed monthly mean wind speed instead of a higher frequency product, focusing on the simulation of monthly mean currents. Considering the objective of generating enduring trends and patterns, monthly means are consistent with the planned temporal resolution. In this application, the assumption is that the currents will achieve a steady state within a month, and using monthly mean wind speed allows for a more computationally efficient technique for modeling these extended currents.

Following the acquisition of the input dataset, a qualitative analysis was conducted to assess data completeness and suitability for subsequent analytical procedures.²⁴ Within the framework of the adaptive mesh model, the parameter values governing the shallow water equations are delineated in Table II. These parameters represent crucial inputs for the computational model, which is pivotal in ensuring the accuracy and reliability of simulations conducted within the designated study area.

C. The initial and lateral boundary conditions

The initial conditions for sea current velocity are set as

$$U = 0, \quad V = 0, \quad \varphi = 0 \text{ (i.e. } \zeta = 0 \text{)}.$$

The lateral boundary condition is

$$U_n|_\Gamma = 0, \quad V_n|_\Gamma = 0,$$

where Γ is the lateral boundary and n represents the normal exterior direction Γ . The component of the current velocity along the outward normal to the boundary is set to zero, i.e., $U = 0$ at the eastward boundary and $V = 0$ at the northward boundary. The preliminary boundary and grid representation of the GoT are depicted in Fig. 3.

D. Splitting methods

The adaptive mesh equations are subdivided into three processes to optimize computational efficiency. These processes encompass the adjustment process, the development process, and the dissipation process. This strategic partitioning facilitates the targeted allocation of computational resources, streamlining the overall numerical solution and enhancing the performance of the adaptive mesh methodology, as shown in Fig. 4.

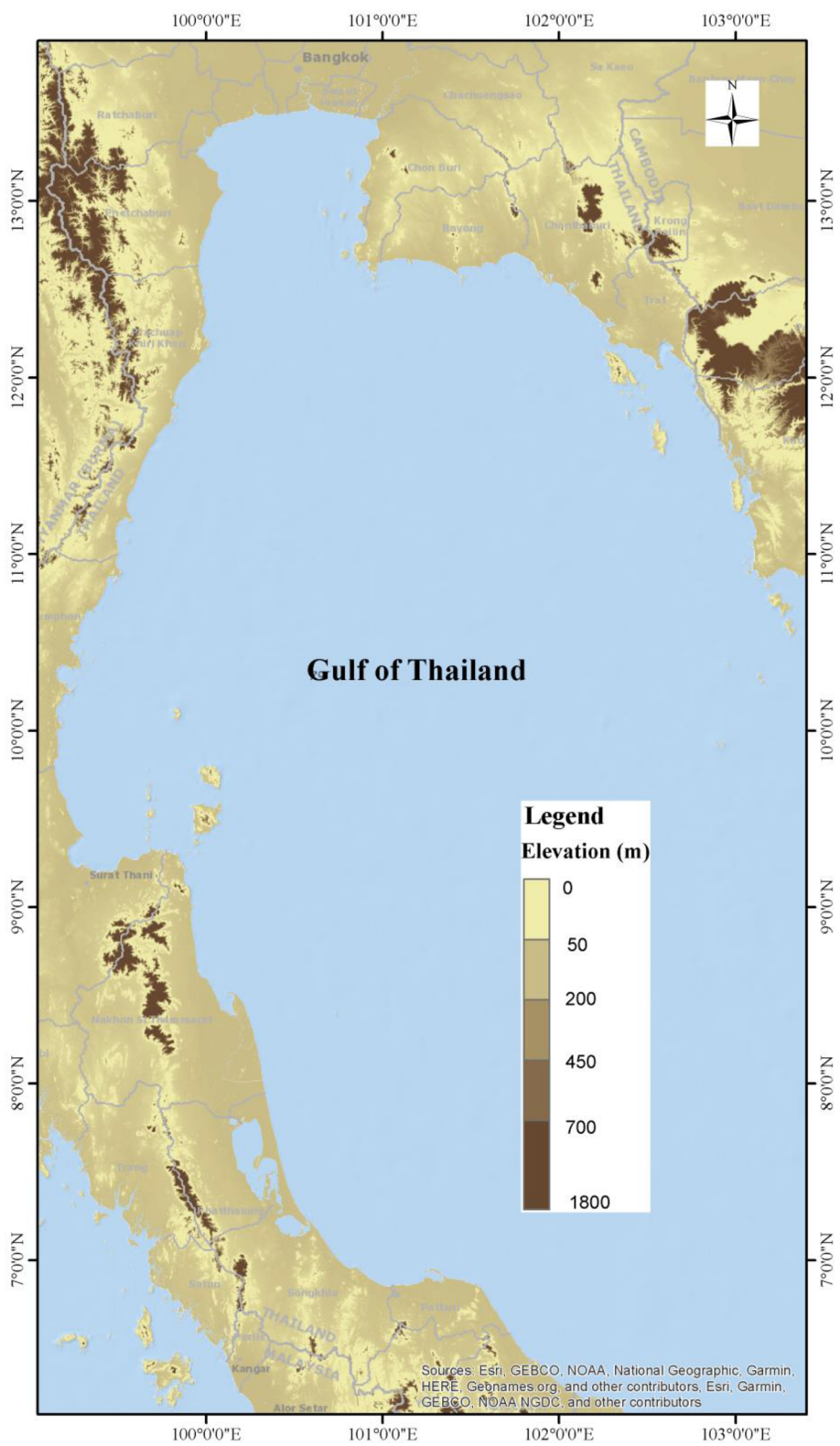


FIG. 1. The Gulf of Thailand (GoT) and Andaman Sea (AS).

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TABLE I. Description of meteorological data.

Data	Description	Source
Wind at 10 m	Reanalysis data, resolution $0.25 \times 0.25^\circ$, type: hourly data. August 1, 2020–2021	Climate data store (CDS)
Surface pressure	Reanalysis data, resolution $0.25 \times 0.25^\circ$, type: hourly data. August 1, 2020–2021	Climate data store (CDS)
Topography	Satellite data, resolution $0.004 \times 0.004^\circ$	General bathymetry chart of the ocean (GEBCO)

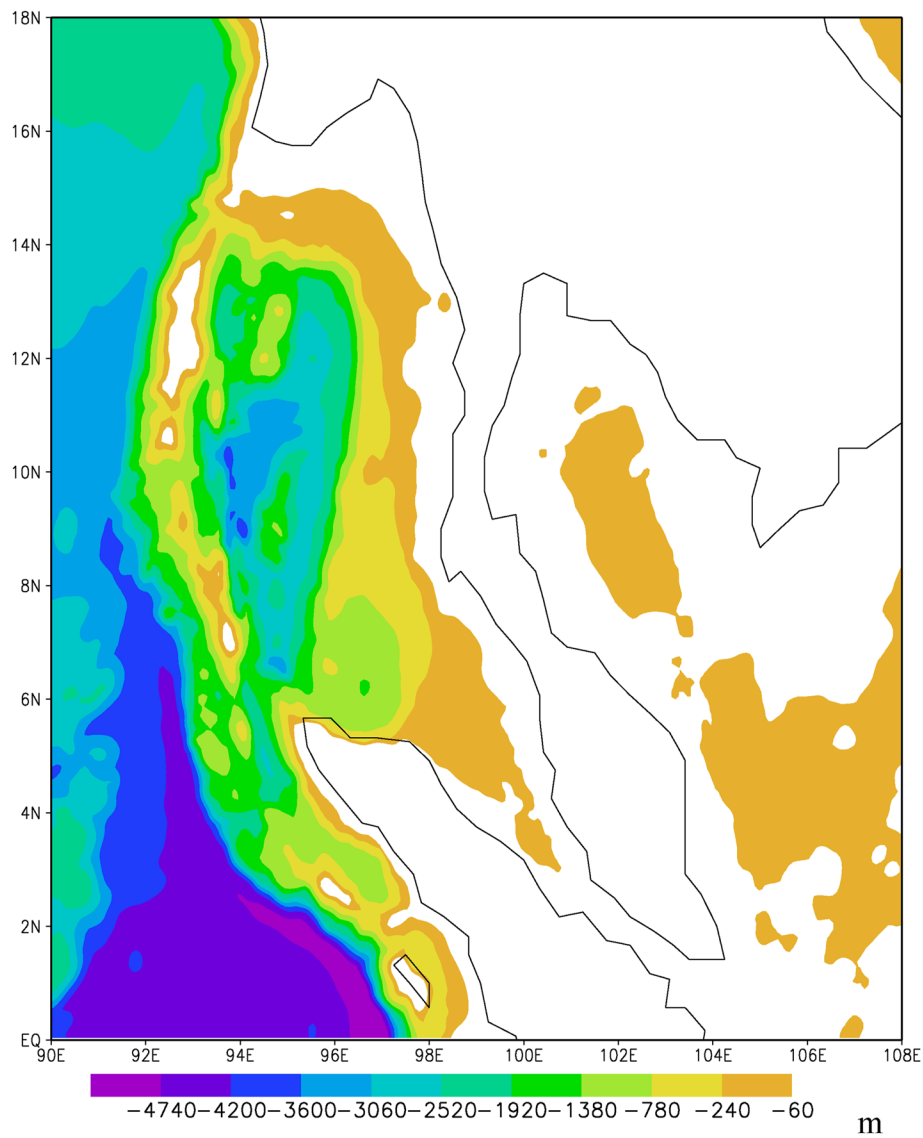


FIG. 2. Bottom topography of the study area.

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TABLE II. The values of parameters used in the model.

Parameter	Symbol	Value	Units
Density of water	ρ	1025	kg/m ³
Density of air	ρ_a	1.226	kg/m ³
Gravity	g	9.806	m/s ²
Angular speed of Earth's rotation	ω	7.292×10^{-5}	rad/s
Earth's radius	r	6.371×10^8	cm
Coefficient of the bottom friction	r_b	3.3×10^{-3}	...
Lateral eddy viscosity coefficient	A_m	1×10^7	...
Coefficient of the wind stress	C_D	1.0×10^{-3}	($ W < 7$ m/s)
		$0.1154 W + 0.1923 \times 10^{-3}$	($7 \leq W < 20$)
		2.5×10^{-3}	($ W \geq 20$ m/s)

1. Adjustment process

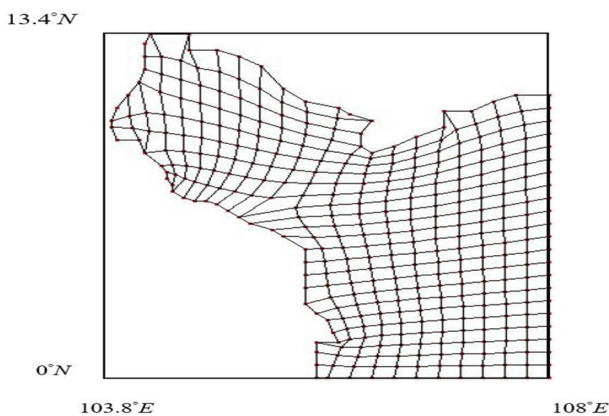
$$\begin{aligned}
 F^I &= F^n - \Delta t_1 \cdot A_1(F^I), \\
 U_{i+\frac{1}{2},j}^I &= U_{i+\frac{1}{2},j}^n + \Delta t_1 \cdot \left((fV)_{i+\frac{1}{2},j} - P_x(\varphi)_{i+\frac{1}{2},j} \right)^I, \\
 V_{i,j+\frac{1}{2}}^I &= V_{i,j+\frac{1}{2}}^n - \Delta t_1 \cdot \left((fU)_{i,j+\frac{1}{2}} - P_y(\varphi)_{i,j+\frac{1}{2}} \right)^I, \\
 \varphi_{i,j+\frac{1}{2}}^I &= \varphi_{i,j+\frac{1}{2}}^n - \Delta t_1 \cdot \left(D(U, V)_{ij} \right)^I,
 \end{aligned}$$

where $\Delta t_1 = 6$ s and $I = n + \frac{1}{3}$.

2. Development process

$$\begin{aligned}
 F^{II} &= F^I - \Delta t_2 \cdot A_2(F^{II}), \\
 U_{i+\frac{1}{2},j}^{II} &= U_{i+\frac{1}{2},j}^I - \Delta t_2 \cdot (L(U))_{i+\frac{1}{2},j}^I, \\
 \varphi_{i,j}^{II} &= \varphi_{i,j}^I,
 \end{aligned}$$

where $\Delta t_2 = 30$ s and $II = n + \frac{2}{3}$.

**FIG. 3.** Study domain in the GoT.

3. Dissipation process

$$\begin{aligned}
 F^{III} &= F^{II} - \Delta t_3 \cdot A_1(F^{III}), \\
 U_{i+\frac{1}{2},j}^{III} &= U_{i+\frac{1}{2},j}^{II} + \Delta t_3 \cdot \left(-(kU)_{i+\frac{1}{2},j} - \frac{\Phi}{\rho} \left(\frac{\partial P_a}{\partial x} \right)_{i+\frac{1}{2},j} + (\Phi F_{sx})_{i+\frac{1}{2},j} \right)^{III}, \\
 V_{i,j+\frac{1}{2}}^{III} &= V_{i,j+\frac{1}{2}}^{II} + \Delta t_3 \cdot \left(-(kV)_{i,j+\frac{1}{2}} - \frac{\Phi}{\rho} \left(\frac{\partial P_a}{\partial y} \right)_{i,j+\frac{1}{2}} + (\Phi F_{sy})_{i,j+\frac{1}{2}} \right)^{III}, \\
 \varphi_{i,j}^{III} &= \varphi_{i,j}^{II},
 \end{aligned}$$

where $\Delta t_3 = 30$ s and $III = n + 1$.

4. Steps of calculations in the modeling

The initial data in the model are the mean current (u and v) and the sea surface elevation (ζ). However, we do not have these values from observation, so the initial data can be prepared as follows: The overall methodology adopted in this research is presented in Fig. 4.

- Step 1: Set grid size $\Delta\xi = \Delta\eta = 55.5$ km.
- Step 2: Use wind stress $\tau_{sx,i,j}$ and $\tau_{sy,i,j}$ from monthly wind stress.
- Step 3: Set initial conditions $U_{i+\frac{1}{2},j} = 0$, $V_{i,j+\frac{1}{2}} = 0$, and $\varphi_{i,j} = 0$.
- Step 4: Set boundary conditions $U_{\frac{1}{2},j} = U_{i-\frac{1}{2},j} = V_{i,\frac{1}{2}} = V_{i,j-\frac{1}{2}} = 0$.
- Step 5: Iteration $it = 0$ and hour = 0.
- Step 6: Use $\Delta t_1 = 6$ s, and run the adjustment process 5 times.
- Step 7: Use $\Delta t_2 = 30$ s, and run the development process 1 time.
- Step 8: Use $\Delta t_3 = 30$ s, and run the dissipation process 1 time.
- Step 9: Iteration $it = it + 1$.
- Step 10: If $(it \bmod 120) = 0$, then hour = hour + 1.
- Step 11: If hour = 168, then go to step 13. Otherwise, go to step 7.
- Step 12: Compute ζu and v .
- Step 13: Stop.

E. Methodological framework for the formulation of adaptive mesh

The fundamental concept underlying the generation method lies in applying the variation principle. By introducing a variation that gauges specific attributes, such as smoothness, orthogonality, and the distribution of grid points, the method seeks to minimize

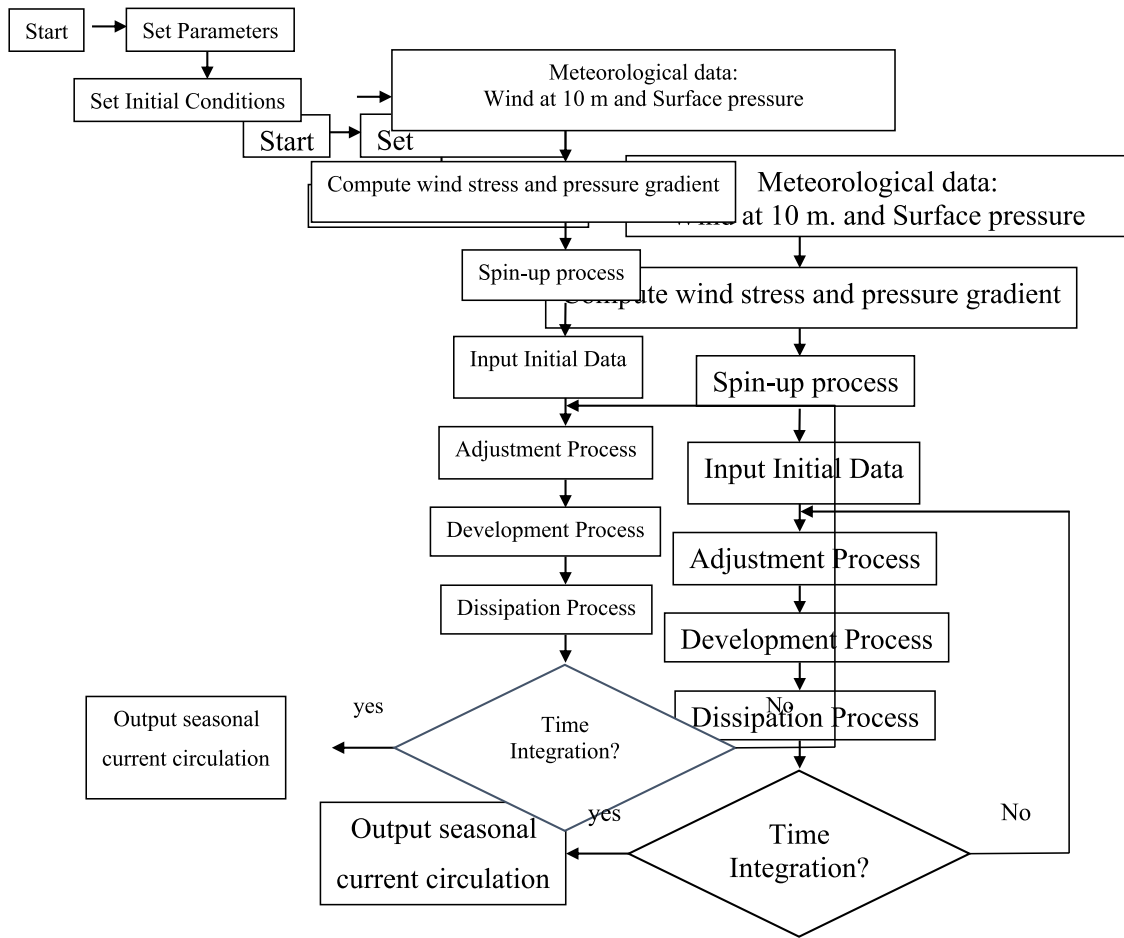


FIG. 4. Overall methodology adopted for monthly mean currents in the Gulf of Thailand.

this variation. Consequently, the Euler equation encapsulates the solution to this minimization process. This approach ensures the generation of optimal grids that align with the desired characteristics, facilitating the attainment of computational solutions with enhanced precision and efficiency.

For any variation, I ,

$$I(x, y) = \iint_{D'} F(x_\xi, x_\eta, y_\xi, y_\eta, x, y) d\xi d\eta,$$

where x, y is the function of ξ, η .

The purpose is to find $x(\xi, \eta), y(\xi, \eta)$, which minimizes the integral function $I(x, y)$.

The minimizing of $I(x, y)$ leads to the following Euler equations:

$$\begin{aligned} -\frac{\partial}{\partial \xi} \left(\frac{\partial F}{\partial x_\xi} \right) - \frac{\partial}{\partial \eta} \left(\frac{\partial F}{\partial x_\eta} \right) + \frac{\partial F}{\partial x} &= 0, \\ -\frac{\partial}{\partial \xi} \left(\frac{\partial F}{\partial y_\xi} \right) - \frac{\partial}{\partial \eta} \left(\frac{\partial F}{\partial y_\eta} \right) + \frac{\partial F}{\partial y} &= 0. \end{aligned}$$

The global smoothness of the mapping is measured by

$$I_x = \iint_D ((\nabla \xi)^2 + (\nabla \eta)^2) dx dy = \iint_D (\xi_x^2 + \xi_y^2 + \eta_x^2 + \eta_y^2) dx dy$$

since

$$\iint_D F(x, y) dx dy = \iint_{D'} F(x(\xi, \eta), y(\xi, \eta)) d\xi d\eta.$$

Hence, on the ξ, η plane,

$$I_x = \iint_D \frac{(\xi_x^2 + \xi_y^2 + \eta_x^2 + \eta_y^2)}{J} d\xi d\eta.$$

Minimizing $I_x(x, y)$ leads to the following Euler equations:

$$\begin{aligned} b_{s1}x_{\xi\xi} + b_{s2}x_{\xi\eta} + b_{s3}x_{\eta\eta} + a_{s1}y_{\xi\xi} + a_{s2}y_{\xi\eta} + a_{s3}y_{\eta\eta} &= 0, \\ a_{s1}x_{\xi\xi} + a_{s2}x_{\xi\eta} + a_{s3}x_{\eta\eta} + c_{s1}y_{\xi\xi} + c_{s2}y_{\xi\eta} + c_{s3}y_{\eta\eta} &= 0, \end{aligned}$$

where

$$\begin{aligned} a_{s1} &= -A\alpha, & b_{s1} &= B\alpha, & c_{s1} &= C\alpha, \\ a_{s2} &= 2A\beta, & b_{s2} &= -2B\beta, & c_{s2} &= -2C\beta, \\ a_{s3} &= -A\gamma, & b_{s3} &= B\gamma, & c_{s3} &= C\gamma. \end{aligned}$$

Under the orthogonal coordinate system $\nabla \xi \cdot \nabla \eta = 0$, it is natural to choose the variation measuring orthogonality by $I_o = \iint_D ((\nabla \xi \cdot \nabla \eta)^2) dx dy$.

However, $\nabla \xi \cdot \nabla \eta = 0$ is not guaranteed where every grid line is orthogonal. Therefore, we weigh the area to measure the orthogonality.

Area weighted variation:

$$I'_o = \iint_D ((\nabla \xi \cdot \nabla \eta)^2) J^3 dx dy = \iint_{D'} (x_\xi x_\eta + y_\xi y_\eta)^2 d\xi d\eta.$$

The minimizing of $I'_o(x, y)$ leads to the following Euler equations:

$$\begin{aligned} b_{o1}x_{\xi\xi} + b_{o2}x_{\xi\eta} + b_{o3}x_{\eta\eta} + a_{o1}y_{\xi\xi} + a_{o2}y_{\xi\eta} + a_{o3}y_{\eta\eta} &= 0, \\ a_{o1}x_{\xi\xi} + a_{o2}x_{\xi\eta} + a_{o3}x_{\eta\eta} + c_{o1}y_{\xi\xi} + c_{o2}y_{\xi\eta} + c_{o3}y_{\eta\eta} &= 0, \end{aligned}$$

where

$$\begin{aligned} a_{o1} &= x_\eta y_\eta, & b_{o1} &= (x_\eta)^2, & c_{o1} &= (y_\eta)^2, \\ a_{o2} &= x_\xi y_\eta + x_\eta y_\xi, & b_{o2} &= 2(x_\xi y_\eta + x_\eta y_\xi), & c_{o2} &= 2(x_\xi y_\eta + 2x_\eta y_\xi), \\ a_{o3} &= x_\xi y_\xi, & b_{o3} &= (x_\xi)^2, & c_{o3} &= (y_\xi)^2. \end{aligned}$$

The weight area variation is measured by

$$I_w = \iint_D W(x, y) J dx dy = \iint_{D'} W(x(\xi, \eta) + y(\xi, \eta)) J^2 d\xi d\eta.$$

The minimizing of $I_w(x, y)$ leads to the following Euler equations:

$$\begin{aligned} 2W(b_{w1}x_{\xi\xi} + b_{w2}x_{\xi\eta} + b_{w3}x_{\eta\eta} + a_{w1}y_{\xi\xi} \\ + a_{w2}y_{\xi\eta} + a_{w3}y_{\eta\eta}) &= -J^2 \frac{\partial W}{\partial x}, \\ 2W(a_{w1}x_{\xi\xi} + a_{w2}x_{\xi\eta} + a_{w3}x_{\eta\eta} + c_{w1}y_{\xi\xi} \\ + c_{w2}y_{\xi\eta} + c_{w3}y_{\eta\eta}) &= -J^2 \frac{\partial W}{\partial y}, \end{aligned}$$

where

$$\begin{aligned} a_{w1} &= -x_\eta y_\eta, & b_{w1} &= (y_\eta)^2, & c_{w1} &= (x_\eta)^2, \\ a_{w2} &= x_\xi y_\eta + x_\eta y_\xi, & b_{w2} &= -2y_\xi y_\eta, & c_{w2} &= -2x_\xi x_\eta, \\ a_{w3} &= -x_\xi y_\xi, & b_{w3} &= (y_\xi)^2, & c_{w3} &= (x_\xi)^2. \end{aligned}$$

This study has obtained three variations, generating three curvilinear coordinates with different properties. If we add these three variations together, we get

$$I = I_s + \lambda_w I_w + \lambda_o I'_o \quad (\lambda_w, \lambda_o \geq 0).$$

Minimizing $I(x, y)$ leads to the following Euler equations:

$$\begin{aligned} b_1x_{\xi\xi} + b_2x_{\xi\eta} + b_3x_{\eta\eta} + a_1y_{\xi\xi} + a_2y_{\xi\eta} + a_3y_{\eta\eta} &= -\lambda_w J^2 \frac{\partial W}{\partial x}, \\ a_1x_{\xi\xi} + a_2x_{\xi\eta} + a_3x_{\eta\eta} + c_1y_{\xi\xi} + c_2y_{\xi\eta} + c_3y_{\eta\eta} &= -\lambda_w J^2 \frac{\partial W}{\partial y}, \end{aligned}$$

where

$$\begin{aligned} a_i &= a_{si} + \lambda_w W a_{wi} + \lambda_o a_{oi}, \\ b_i &= b_{si} + \lambda_w W b_{wi} + \lambda_o b_{oi}, \\ c_i &= c_{si} + \lambda_w W c_{wi} + \lambda_o c_{oi}. \end{aligned}$$

1. Numerical solution

The finite difference approximations to the Euler equation are solved by the Jacobi iteration. Then for (ξ, η) , there are continuous variables at the nodes of the computation mesh, which is each node's symbol (ξ, η) . The difference equations approximate the value of the derivatives at the nodes,

$$\begin{aligned} x_\xi &\simeq \frac{1}{2}(x_{i+1,j} - x_{i-1,j}), \\ x_\eta &\simeq \frac{1}{2}(x_{i,j+1} - x_{i,j-1}), \end{aligned}$$

and by the second derivative,

$$\begin{aligned} x_{\xi\xi} &= x_{i+1,j} - 2x_{i,j} + x_{i-1,j}, \\ x_{\xi\eta} &= \frac{1}{4}(x_{i+1,j+1} + x_{i-1,j-1} - x_{i+1,j-1} - x_{i-1,j+1}), \\ x_{\eta\eta} &= x_{i,j+1} - 2x_{i,j} + x_{i,j-1}. \end{aligned}$$

This is similar for $y(\xi, \eta)$.

From many examples, it can be seen that the mesh was affected by $I_s, \lambda_w I_w, \lambda_o I'_o$.

F. Formulation of adaptive mesh

The principal generation of adaptive mesh is the coordinate transformation $x(\xi, \eta)$ and $y(\xi, \eta)$, which maps a rectangular domain D' of the $\xi - \eta$ plane into the physical domain D of the $x - y$ plane so that it is a one-on-one mapping between the boundaries of the rectangular and calculation domain and between the inner parts of the domain, as mentioned by Hagen (Fig. 5)². The finite difference method may be implemented directly within the parameter plane, obviating the need to account for the configuration of the physical boundary. This approach allows for calculating a fixed, equidistant, and uniformly rectangular grid system, irrespective of the dynamic nature of the computational boundary. The grid's boundary aligns precisely with the corresponding physical domain boundaries, ensuring computational accuracy and efficiency in scenarios characterized by temporal variability.

To transform the equations under the physical plane coordinates (x, y, t) into the parameter plane coordinates (ξ, η, τ) ,

$$x = x(\xi, \eta, \tau), \quad y = y(\xi, \eta, \tau), \quad t = \tau.$$

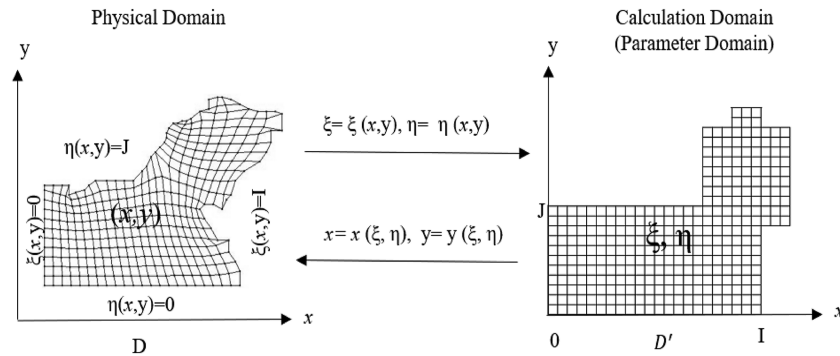


FIG. 5. Sketch map of coordinate transformation.

The following relationships of coordinate transformation are needed, where $F(x, y, t)$ is any function on physical plane coordinates (x, y, t) ,

$$\begin{aligned} \left(\frac{\partial F}{\partial x} \right)_{(x,y,t)} &= \frac{1}{J} \left[y_\eta \left(\frac{\partial F}{\partial \xi} \right)_{(\xi,\eta,\tau)} - y_\xi \left(\frac{\partial F}{\partial \eta} \right)_{(\xi,\eta,\tau)} \right], \\ \left(\frac{\partial F}{\partial y} \right)_{(x,y,t)} &= \frac{1}{J} \left[-x_\eta \left(\frac{\partial F}{\partial \xi} \right)_{(\xi,\eta,\tau)} + x_\xi \left(\frac{\partial F}{\partial \eta} \right)_{(\xi,\eta,\tau)} \right], \\ \left(\frac{\partial F}{\partial t} \right)_{(x,y,t)} &= \left(\frac{\partial F}{\partial \tau} \right)_{(\xi,\eta,\tau)} - x_\tau \left(\frac{\partial F}{\partial x} \right)_{(x,y,t)} - y_\tau \left(\frac{\partial F}{\partial y} \right)_{(x,y,t)}, \end{aligned}$$

where $J = x_\xi y_\eta - y_\xi x_\eta$ is the Jacobian of the coordinate transformation.

Then,

$$\begin{aligned} \left(\frac{\partial F}{\partial t} \right)_{(x,y,t)} &= \left(\frac{\partial F}{\partial \tau} \right)_{(\xi,\eta,\tau)} - \frac{1}{J} (y_\xi x_\tau - x_\eta y_\tau) \left(\frac{\partial F}{\partial \xi} \right)_{(\xi,\eta,\tau)} \\ &\quad - \frac{1}{J} (-y_\xi x_\tau + x_\eta y_\tau) \left(\frac{\partial F}{\partial \eta} \right)_{(\xi,\eta,\tau)}. \end{aligned}$$

As the coordinates $x = x(\xi, \eta, \tau)$, $y = y(\xi, \eta, \tau)$ are a function of time τ , the Jacobian term J is also a function of time $x = x(\xi, \eta, \tau)$, $y = y(\xi, \eta, \tau)$, $t = \tau$,

$$\begin{aligned} \frac{\partial J}{\partial \tau} &= x_{\xi\tau} y_\eta + x_\xi y_{\eta\tau} - y_{\xi\tau} x_\eta - y_\xi x_{\eta\tau} \\ &= x_{\tau\xi} y_\eta - x_{\tau\eta} y_\xi + x_\xi y_{\tau\eta} - x_\eta y_{\tau\xi} \\ &= (x_\tau y_\eta)_\xi - (x_\tau y_\xi)_\eta + (x_\xi y_\tau)_\eta - (x_\eta y_\tau)_\xi \\ &= (y_\eta x_\tau - x_\eta y_\tau)_\xi + (-y_\xi x_\tau + x_\xi y_\tau)_\eta. \end{aligned}$$

Bomers *et al.*³ introduced grid generation by solving Poisson's boundary value problem,

$$\frac{\partial^2 \xi}{\partial x^2} + \frac{\partial^2 \xi}{\partial y^2} = P(\xi, \eta), \quad \frac{\partial^2 \eta}{\partial x^2} + \frac{\partial^2 \eta}{\partial y^2} = Q(\xi, \eta),$$

where $P(\xi, \eta)$, $Q(\xi, \eta)$ were used to adjust the density of grid points. P can be chosen as

$$\begin{aligned} P(\xi, \eta) &= -\sum_{i=1}^n a_i \text{sign}(\xi - \xi_i) \exp(-c_i |\xi - \xi_i|) - \sum_{j=1}^n b_j \text{sign}(\xi - \xi_j) \\ &\quad \times \exp\left(-d_j ((\xi - \xi_j)^2 + (\eta - \eta_j)^2)^{\frac{1}{2}}\right), \end{aligned}$$

where the choice of a_i, c_i is to make the line $\xi = c$ concentrate to the line $\xi = \xi_i$, and the choice b_j, d_j is to let the line $\xi = c$ concentrate to the point $(\xi_i, \eta_j) \cdot Q(\xi, \eta)$ of similar form.

The inverse transformation of the corresponding transformation (3) is

$$\begin{aligned} \alpha x_{\xi\xi} - 2\beta x_{\xi\eta} + \gamma x_{\eta\eta} &= -J^2 (Px_\xi + Qx_\eta), \\ \alpha y_{\xi\xi} - 2\beta y_{\xi\eta} + \gamma y_{\eta\eta} &= -J^2 (Py_\xi + Qy_\eta), \end{aligned}$$

where α, β , and γ are the quadratic terms of the first derivative of x, y and ξ, η , and J is the Jacobian term of the transformation $J = x_\xi y_\eta - x_\eta y_\xi$.

The mesh generation process unfolds systematically, encompassing the following key steps:

1. Adaptation to the boundary of the calculation domain: The initial step involves the adjustment of the mesh to conform precisely to the boundaries delineating the calculation domain. It ensures that the computational grid aligns accurately with the physical domain under consideration.
2. Subsequently, the mesh undergoes adaptation based on the spatial distribution of pertinent physical variables. This step optimizes the grid to accommodate variations in the distribution of these variables, enhancing computational accuracy.
3. A critical aspect of the mesh generation process is the preservation of smoothness and orthogonality in the arrangement of coordinate lines. This step aims to maintain grid integrity, promoting computational stability and precision.
4. Finally, the finite difference method is utilized on a parameter plane. The final phase involves the application of the finite difference method on a parameter plane. This plane employs an equidistant, uniformly rectangular mesh coordinate system. Importantly, this approach obviates the need to

explicitly consider the shape of the boundary and the distribution of physical variables, simplifying the computational process while maintaining accuracy.

C. Adaptive mesh examples I

The adaptive mesh is specifically tailored to the boundaries of the Geographic Information System (GIS) data, delineating a domain constrained within latitudes 0° N to 13.4° N and longitudes 103.8° E to 108° E, as illustrated in Figs. 6(a) and 6(b). This mesh adaptation ensures precise alignment with the geographical features and spatial coordinates defined by the specified latitude and longitude ranges, facilitating accurate representation and analysis within the designated region ($\lambda_w = 5, \lambda_o = W(x, y) = 1$).

1. Example II

The adaptive mesh configuration is designed to conform to the boundaries of the Andaman Sea, encapsulated within latitudes 0° N to 19° N and longitudes 94° E to 108° E, as depicted in Figs. 7(a) and 7(b). This adaptive mesh ensures precise representation and

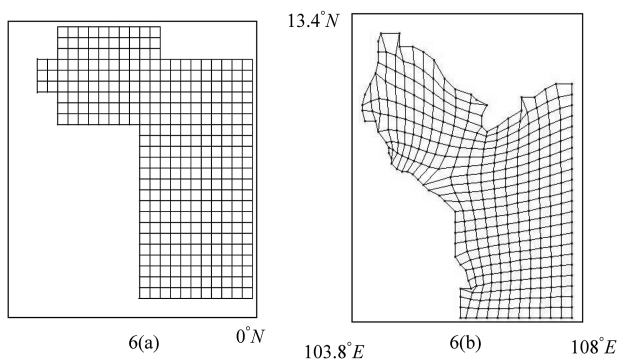


FIG. 6. (a) Calculation domain and (b) geographical domain in the GoT.

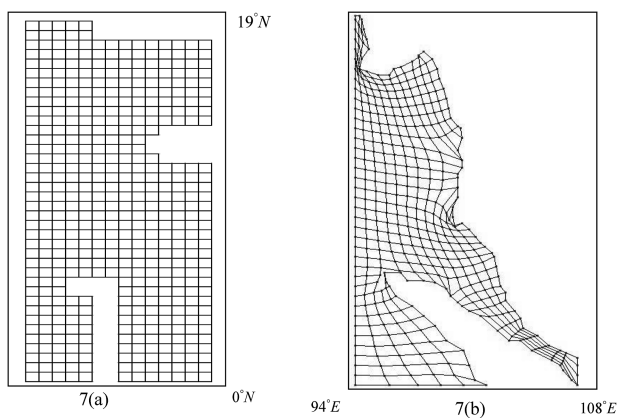


FIG. 7. (a) Calculation domain and (b) geographical domain in the AM sea.

computational accuracy within the specified geographical domain, aligning with the defined latitude and longitude ranges associated with the Andaman Sea region. The mesh adaptation facilitates an effective modeling and analysis framework for the targeted area ($\lambda_w = 5, \lambda_o = W(x, y) = 1$).

III. RESULTS FOR THE MONTHLY MEAN CURRENTS OF THE GULF OF THAILAND

The utilization of adaptive mesh grids and conventional regular meshes for modeling the ocean dynamics in the GoT demands precise examination. The simulated monthly mean currents within the GoT manifest in three distinct groupings, described by the influence of the northeast monsoon (November to January), the southwest monsoon (April to August), and transitional months (February to March and September to October), as demonstrated in Figs. 8–15. First, the conventional regular meshes shown in Figs. 8–11 reproduce the monthly mean current patterns. Adaptive Mesh (AM) grids, featuring dynamic spatial resolution adjustments, appropriately capture intricate flow patterns, enhancing precision through targeted concentration of computational resources. However, this adaptability necessitates heightened computational demands. Conventional regular meshes, in contrast, preserve computational efficiency via uniform resolution, potentially compromising accuracy in locales demanding fine-scale fidelity. The adaptability of AM grids is underscored by their adept alignment of grid points with the actual domain boundary, a pivotal factor in ensuring coastal modeling accuracy. Model validation attests to the congruence between simulated monthly mean current circulations and empirical data, validating the efficacy of the adaptive mesh paradigm. Ultimately, the selection between adaptive and regular meshes pivots on the nuanced equilibrium among precision, computational efficiency, and model adaptability.

Under the impact of the northeast monsoon (between November and January), prevailing wind stress in the GoT predominantly directs southward. The seasonal current simulation in the GoT (Fig. 12) illustrates a southward flow originating from the SCS, which infiltrates the GoT, moves northward, executes a subsequent westward turn, and ultimately discharges into the south. Transitioning into the southwest monsoon phase from the Indian Ocean from February to March, the wind over the Andaman Sea flows from west to east and undergoes a clockwise rotation toward the west. Concurrently, the wind over the GoT flows northward from the east. Predominantly, the wind stress direction converges from the south into the ground. The simulated seasonal current pattern in the GoT (Fig. 13) depicts a flow from Malaysia along the Southern Thailand coastline, proceeding northward to the upper GoT, executing a clockwise rotation within the central region, and exiting into the SCS. As the southwest monsoon prevails (from April to August), the wind stress primarily directs northeastward from the west over the GoT. The simulated seasonal current dynamics (Fig. 14) showcase coastal currents with velocities exceeding 0.25 m/s moving northward to the upper GoT, accompanied by a minor clockwise rotation with speeds around 0.15 m/s in the central Gulf region. This flow integrates southward with contributions from Malaysia, ultimately discharging into the SCS. Transitioning from the southwest monsoon to the onset of the northeast monsoon in September to October, wind stress reverses, flowing west to east over the GoT.

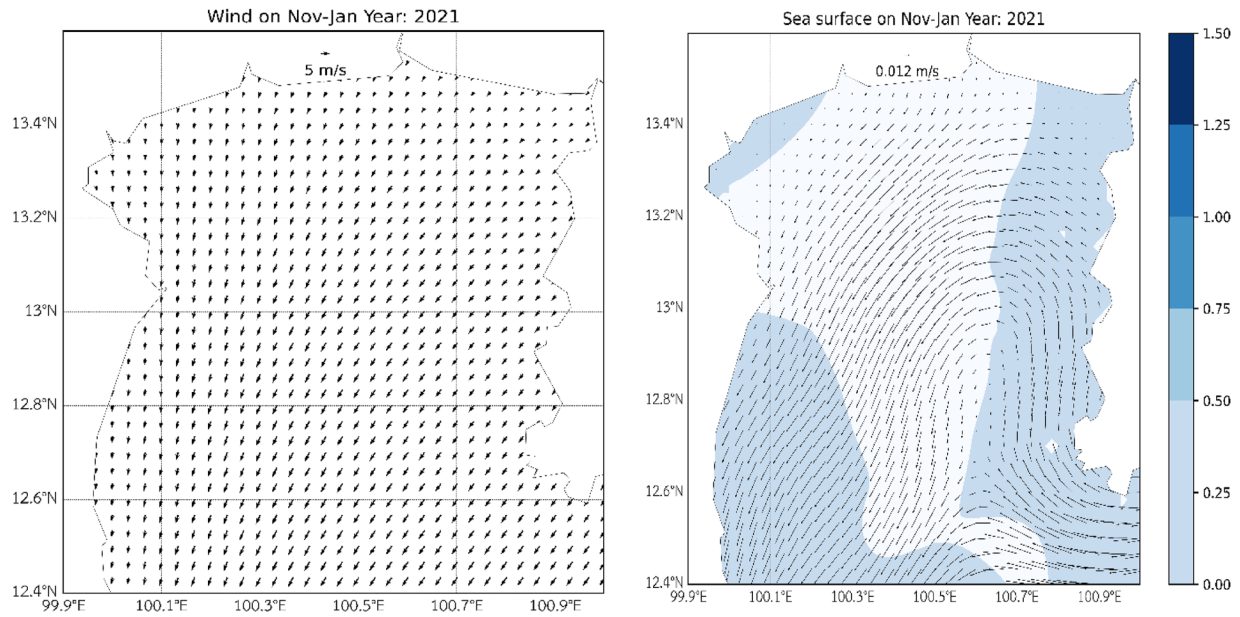


FIG. 8. Circulation of the sea surface in November–January (2021) by conventional regular meshes: (a) wind stress (m/s); (b) current circulation (m/s).

The simulated seasonal current patterns in the GoT (Fig. 15) reveal southward currents turning westward along the coast, with inflowing currents from the SCS reaching the upper GoT and diverting eastward along Rayong. A subtle counterclockwise monthly mean current, with velocities below 0.1 m/s, is observed near the central

GoT, followed by an outflow into the SCS. The agreement between the simulated monthly mean current circulations and observations is noteworthy. In addition, the alignment of adaptive mesh grid points at the boundary coincides seamlessly with the physical domain boundary.

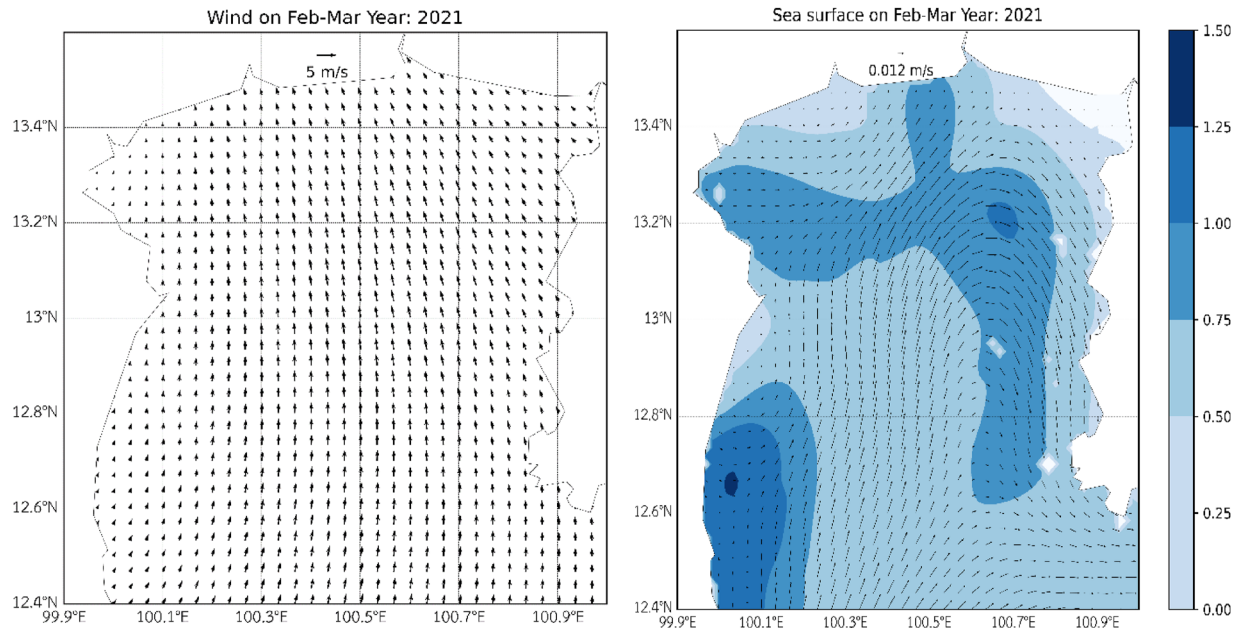


FIG. 9. Circulation of the sea surface in February–March (2021) by conventional regular meshes: (a) wind stress (m/s); (b) current circulation (m/s).

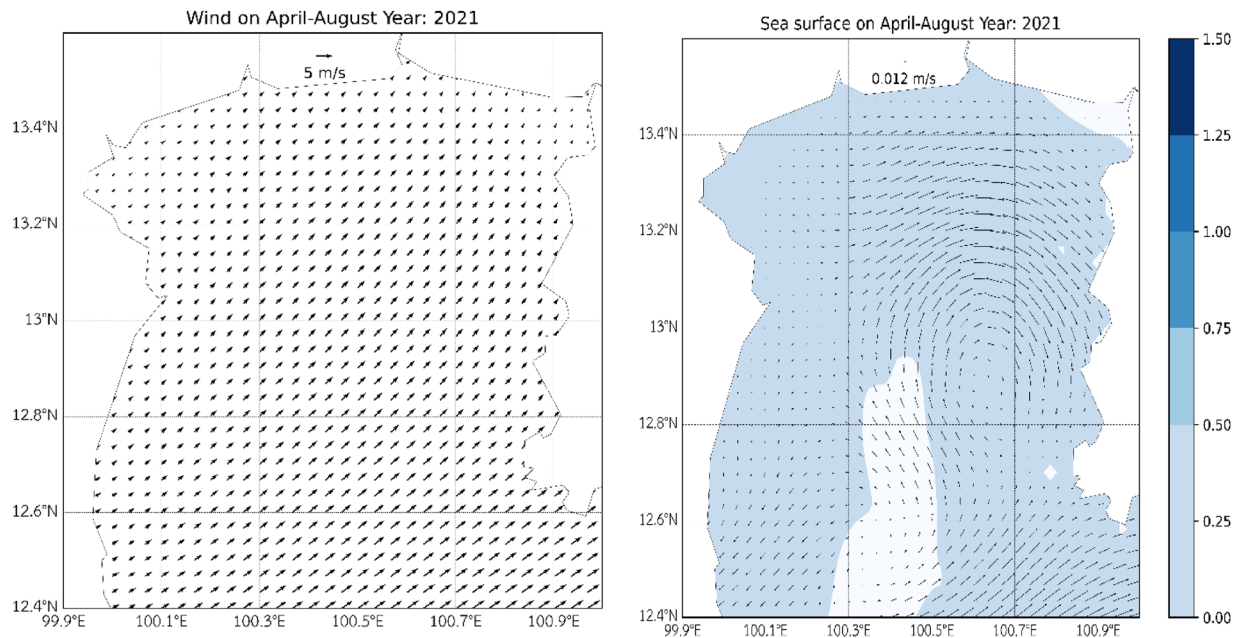


FIG. 10. Circulation of the sea surface in April–August (2021) by conventional regular meshes: (a) wind stress (m/s); (b) current circulation (m/s).

The simulated monthly mean current circulations exhibit a commendable concordance with observational data. Using AM coordinates is a highly effective tool for solving partial differential equations across arbitrary domains. Unlike conventional

regular meshes that discretize domains uniformly, the AM employs a nonuniform curvilinear structure capable of automatic adjustment to specific solution features. This adaptability accommodates the shape of the calculation domain and the spatial distribution

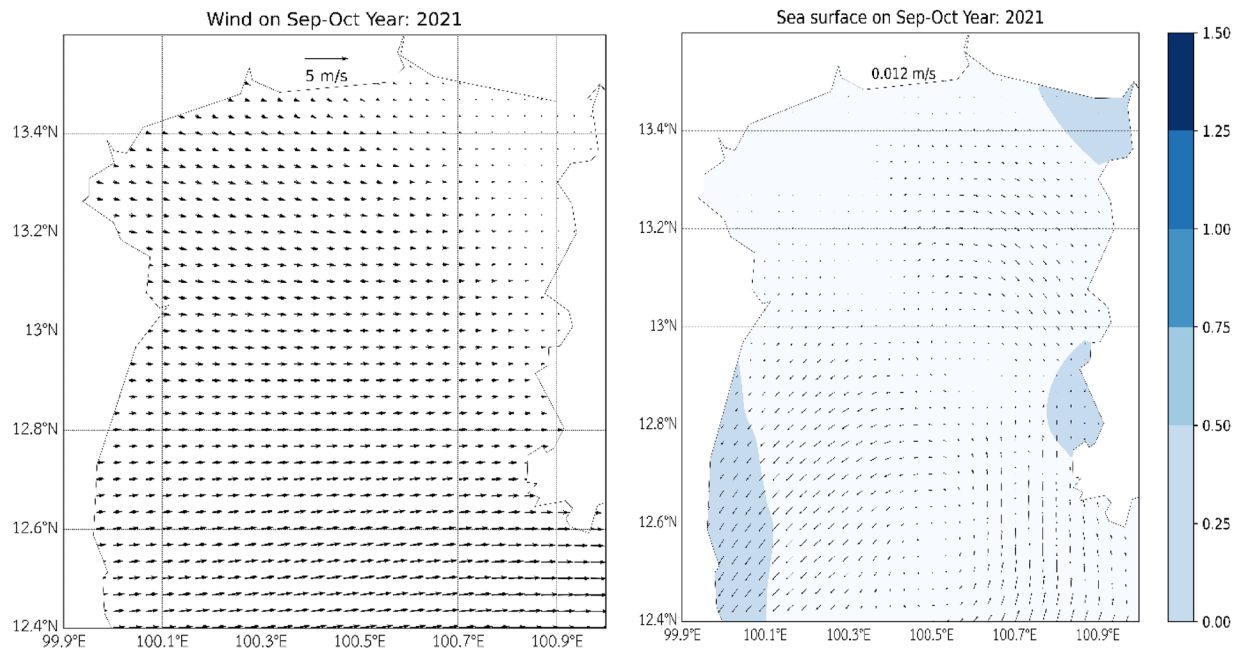


FIG. 11. Circulation of the sea surface in September–October (2021) by conventional regular meshes: (a) wind stress (m/s); (b) current circulation (m/s).

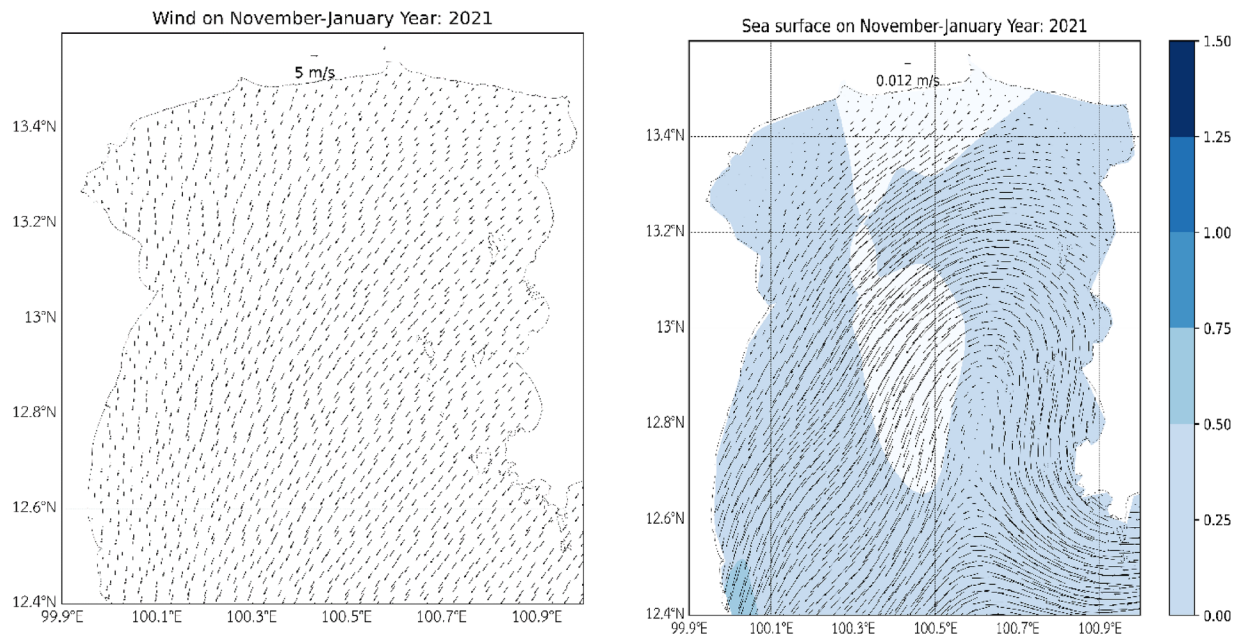


FIG. 12. Circulation of the sea surface in November–January (2021) by adaptive mesh: (a) wind stress (m/s); (b) current circulation (m/s).

of physical variables, ensuring that the adaptive mesh grid aligns seamlessly with the physical domain's boundaries. The adaptive mesh coordinate technique was used to calculate the monthly mean currents in the GoT. The study domain spanned from 99° E to

108° E in longitude and 0° N to 13.8° N in latitude for the GoT. Boundary grid points were initially established, the computational domain was designed, and adaptive mesh grids were subsequently generated.

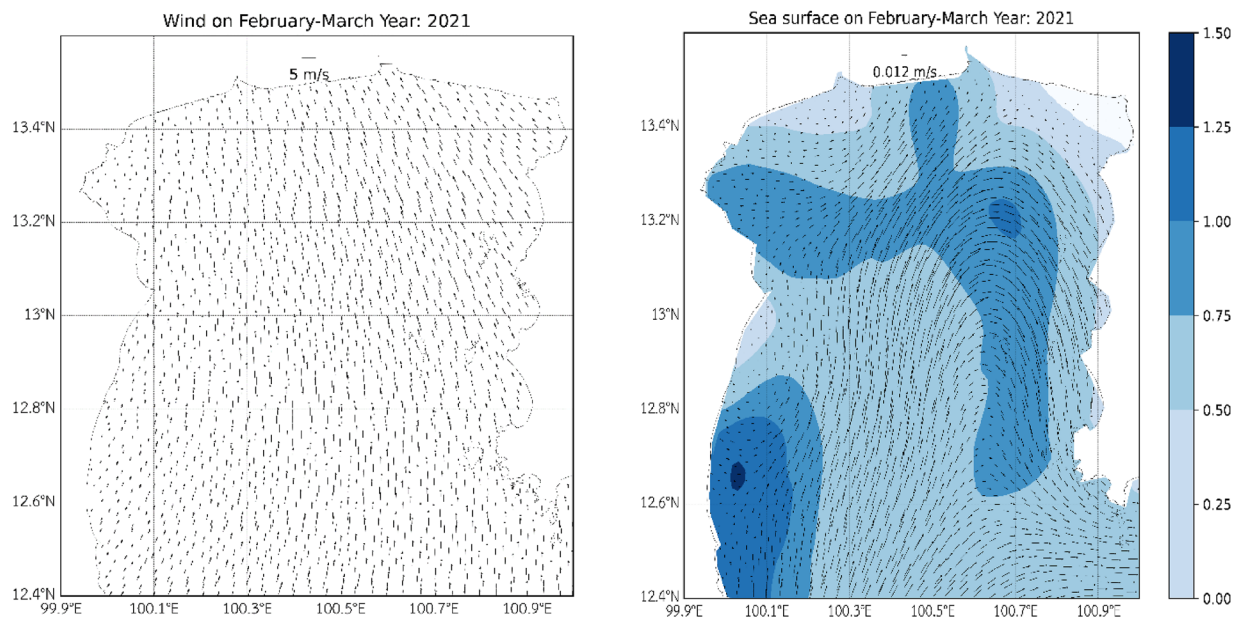


FIG. 13. Circulation of the sea surface in February–March (2021) by adaptive mesh: (a) wind stress (m/s); (b) current circulation (m/s).

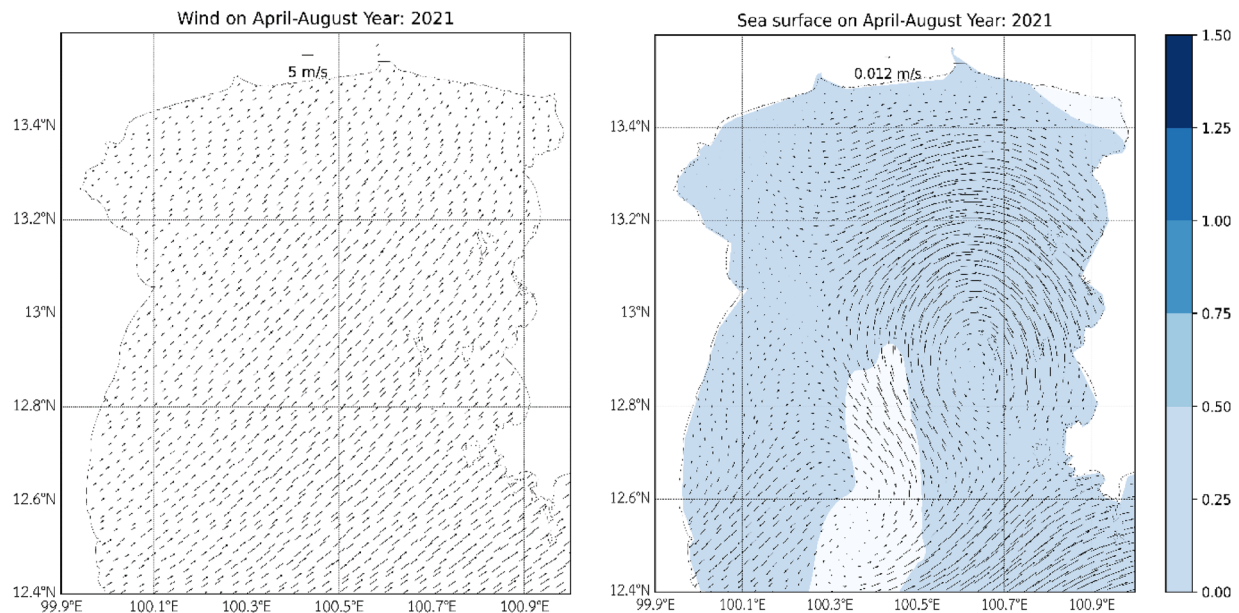


FIG. 14. Circulation of the sea surface in April–August (2021) by adaptive mesh: (a) wind stress (m/s); (b) current circulation (m/s).

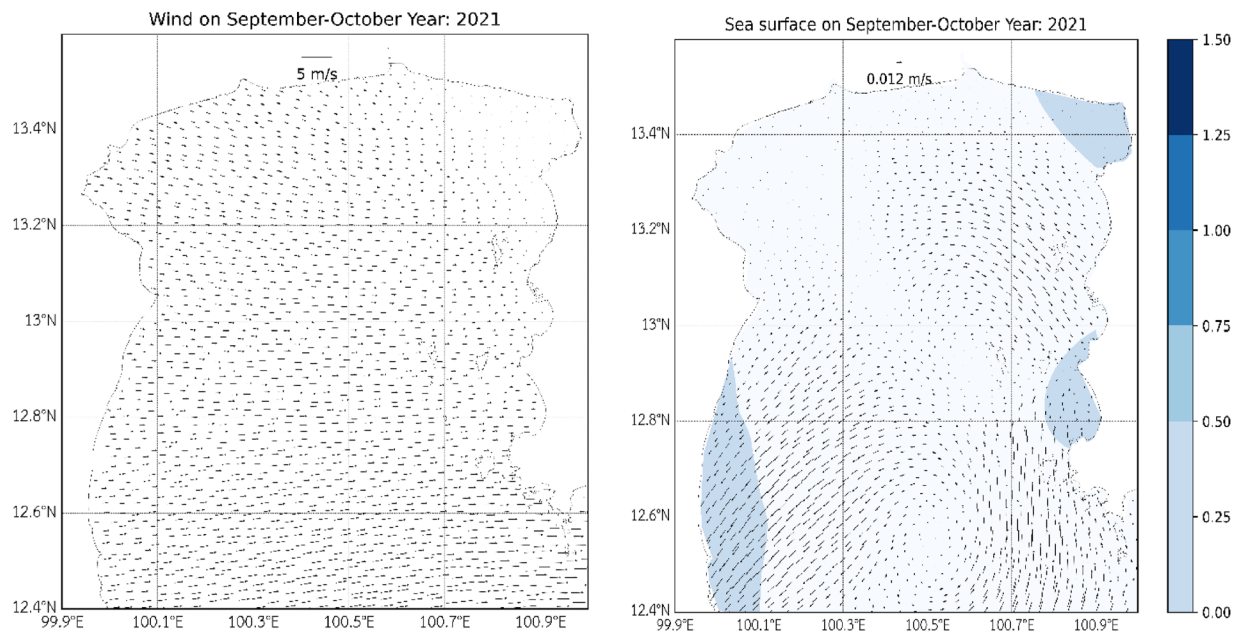


FIG. 15. Circulation of sea surface in September–October (2021) by adaptive mesh: (a) wind stress (m/s); (b) current circulation (m/s).

IV. CONCLUSION AND RECOMMENDATIONS

In this study, we have undertaken the development of an adaptive mesh coordinate model to systematically generate grids within the geographical confines of the GoT and the AS. The objective was to establish boundary-conforming adaptive mesh discretizations for

the shallow water equations, enabling the computation of monthly mean ocean currents in the vicinity of Thailand, particularly focusing on the regions surrounding the GoT and AS. The formulation and application of this adaptive mesh coordinate model represent significant strides forward in oceanographic modeling and simulation. By generating grids that precisely conform to the intricate

coastal boundaries of these regions, we have greatly improved the precision and applicability of numerical simulations. This development is crucial, given the complex coastal features and dynamic oceanic processes present in these areas. The resultant monthly mean current data will offer valuable insights into the dynamic behaviors of the GoT and the AS with potential implications for various scientific and practical applications. These applications include but are not limited to environmental monitoring, navigation, and resource management. The ability to accurately model and predict ocean currents in these regions can aid in the understanding of environmental processes and assist in making informed decisions regarding coastal management and conservation efforts. This research underscores the importance of adaptive mesh coordinate modeling as a powerful tool for simulating complex coastal features and oceanic processes. By utilizing this approach, we can better capture the intricate dynamics of coastal regions, which are often characterized by varying bathymetry, coastline geometry, and other environmental factors. Furthermore, our study opens avenues for future research endeavors. It is recommended that the adaptive mesh coordinate model be further refined and validated against observational data to enhance its accuracy and reliability. This validation process is essential for ensuring that the model accurately represents the real-world conditions of the GoT and the AS.

In addition, future research should explore the incorporation of temporal dynamics to analyze monthly mean ocean currents over various time scales. This temporal analysis will provide a more comprehensive understanding of the underlying processes driving ocean currents in these regions and how they may vary seasonally or in response to external forcings. Moreover, integrating additional environmental factors into the model, such as temperature, salinity, and sediment transport, can further enhance its predictive capabilities. Engaging stakeholders and addressing specific environmental issues will also be crucial for translating the research findings into actionable insights for coastal management and conservation efforts. Scaling up the model to cover larger areas and fostering interdisciplinary collaborations can advance the field of coastal processes analysis. By collaborating with experts from diverse disciplines, such as oceanography, meteorology, and marine biology, we can gain a more holistic understanding of coastal dynamics and their implications for the environment and society.

AUTHOR DECLARATIONS

Conflict of Interest

The authors have no conflicts to disclose.

Author Contributions

Boobphachard Chansawang: Formal analysis (equal); Writing – original draft (equal). **Rahat Zarin:** Data curation (equal); Writing – original draft (equal). **Usa Wannasingha Humphries:** Project administration (equal); Supervision (equal); Writing – review & editing (equal). **Prungchan Wongwises:** Investigation (equal); Methodology (equal). **Muhammad Waqas:** Validation (equal); Visualization (equal). **Angkool Wangwongchai:** Investigation (equal); Software (equal).

DATA AVAILABILITY

The data that support the findings of this study are available from the corresponding author upon reasonable request.

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